

Transportation flow chart for St. Paul (Passenger)

Inputs for VMT calculation

- PMT forecasts from Met Council regional transportation demand model (RTDM)
- Average vehicle occupancy by mode
- MA3T stock turnover and adoption forecasts by powertrain: SI (spark ignition gasoline) vehicles, CI (combustion emissions diesel) vehicles, hybrid-electric vehicles (HEVs), plug-in hybrid-electric vehicles (PHEV), and battery-electric



Equation for VMT calculation with terms defined

$VMT_{SI\ PLDV} = PMT_{PLDV} \times \frac{SI\ Stock}{PLDV\ Stock} \times \frac{1}{AVO}$: $VMT_{SI\ PLDV}$ is total annual VMT by SI vehicles. PMT_{PLDV} is total PMT by passenger light-duty vehicles (PLDV). $SI\ Stock$ is the stock of SI vehicles; $PLDV\ Stock$ is the total stock of PLDV. AVO is the average vehicle occupancy.

Inputs for GHG calcs w assumptions

- Fuel consumption by mode and powertrain
- Direct GHG emissions factor (EF) for fuel.
- Indirect GHG emissions factor for the manufacturing and maintenance of vehicles.



Equation for GHG calculation

$$Direct\ GHG = \frac{VMT}{MPG} \times EF$$

Direct GHG are the emissions from the combustion of fuel for vehicle operations

MPG is the fuel efficiency measured as either miles per gallon (for SI, CI, HEV, and part of PHEV VMT) or miles per kWh (for BEV and part of PHEV VMT)

EF is emission factor. Gasoline and diesel EF are emissions from fuel combustion. Electricity EF is emissions at the grid level, extracted from eGrid data.

$$Indirect\ GHG = SI\ Sales \times EF$$

Indirect emissions are calculated by mode and powertrain type for vehicle sales in each year of analysis. Sales are adjusted in response to user inputs.

Inputs for cost and energy saved

See cost table.



Equation for cost calculation

Cost = Cost per VMT * total VMT
by mode and fuel source

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(Passenger)**

List of interventions	Two-line description in simple language	How calculated
A. Compact urban form		The maximum effect of compact urban form is set at a 25% reduction in auto VMT.
A1. Increase population density by 5% by 2040.	Increasing density decreases travel distances and encourages the use of transit, cycling, and walking. We estimate a 0.4% decrease in auto VMT for a 10% increase in population density by 2040, a 7% increase in walking/cycling VMT for a 10% increase in population density by 2040, and a 7% increase in transit VMT for a 10% increase in population density by 2040.	Applied to basic VMT calculation as additional factor that adjusts total by mode up/down depending on the sign of the change. Population density measured as number of persons per land area.
A2. Increase employment density by 5% by 2040.	Increasing density decreases travel distances and encourages the use of transit, cycling, and walking. We estimate no change in auto VMT for a 10% increase in employment density by 2040, a 0.4% increase in walking/cycling VMT for a 10% increase in employment density by 2040, and a 0.1% increase in transit VMT for a 10% increase in employment density by 2040.	Applied to basic VMT calculation as additional factor that adjusts total by mode up/down depending on the sign of the change. Employment density measured as number of persons per land area.
A3. Increase land use diversity by 5% by 2040.	Increasing land use diversity/mixing decreases travel distances and encourages the use of transit, cycling, and walking. We estimate a 0.9% decrease in auto VMT for a 10% increase in land use diversity by 2040, a 1.5 % increase in walking/cycling VMT for a 10% increase in land use diversity by 2040, and a 1.2% increase in transit VMT for a 10% increase in land use diversity by 2040.	Applied to basic VMT calculation as additional factor that adjusts total by mode up/down depending on the sign of the change. Land use typically quantified using an entropy measure.
A4. Increase fraction of 4-way intersections by 5% by 2040.	Increasing 4-way intersections, and other design changes to increase road connectivity, decreases travel distance by reducing indirect travel. We estimate a 1.2% decrease in auto	Applied to basic VMT calculation as additional factor that adjusts total by mode up/down depending on the sign of the change. Fraction of 4-way intersections can be calculated using OpenStreetMap data and GIS tools.

**Transportation flow chart for St. Paul
(Passenger)**

	VMT for a 10% increase in 4-way intersections by 2040, a 0.6% decrease in walking/cycling VMT for a 10% increase in 4-way intersections by 2040, and a 2.9% increase in transit VMT for a 10% increase in 4-way intersections by 2040. Walking/cycling VMT decreases due to more direct travel while transit VMT increases due to mode shift away from auto travel.	
A5. Increase job accessibility by 5% by 2040.	Increasing job accessibility decreases travel time. We estimate a 2.0% decrease in auto VMT for a 10% increase in job accessibility by 2040, a 0.6% decrease in walking/cycling VMT for a 10% increase in 4-way intersections by 2040, and a 1.3% increase in transit VMT for a 10% increase in job accessibility by 2040.	Applied to basic VMT calculation as additional factor that adjusts total by mode up/down depending on the sign of the change. Job accessibility can be measured as number of jobs within a given travel time (e.g., 30 minute drive), using a gravity measure, or using a utility measure.
A6. Decrease distance to transit by 5% by 2040.	Decreasing distance to transit encourages the use of transit. We estimate a 0.5% decrease in auto VMT for a 10% decrease in the distance to the nearest transit stop by 2040, a 1.5% increase in walking/cycling VMT for a 10% decrease in the distance to the nearest transit stop by 2040, and a 2.9% increase in transit VMT for a 10% decrease in the distance to the nearest transit stop by 2040.	Applied to basic VMT calculation as additional factor that adjusts total by mode up/down depending on the sign of the change. Distance to nearest transit stop can be measured using GTFS or other data sources and GIS tools.
A7. No urban expansion. Increase population density (as a composite measure) by 2% by 2040 relative to BAU.	Changes in density typically happens in combination with changes in land use diversity, design, job accessibility, and distance to transit (i.e., all 5Ds). We estimate a 2.2% decrease in auto VMT for a 10% increase in population density by 2040, a 3.3% increase in walking/cycling VMT for a 10% increase in population density by 2040,	Applied to basic VMT calculation as additional factor that adjusts total by mode up/down depending on the sign of the change. Urban expansion measured as developed land in 2040 relative to base year. Population density can be measured for this developed land area relative to an alternative scenario of no expansion from the base year extent.

**Transportation flow chart for St. Paul
(Passenger)**

	and a 6.2% increase in transit VMT for a 10% increase in population density by 2040.	
B. Vehicle electrification		
B1. Increase private EV sales to 20% HEV, 20% PHEV, and 60% BEV of all vehicle sales by 2040.	Electric vehicles (EVs) are more energy efficient than internal combustion engine (ICE) vehicles and will further reduce emissions as the grid decarbonizes. The model assumes 6% of vehicles are hybrid electric (HEVs) with fuel economy of 65.6 mi/gal, 4% of vehicles are plug-in hybrid electric (PHEVs) with fuel economy of 57.1 mi/gal in gasoline-mode and 3.1 miles/kWh in electric-mode, and 6% of vehicles are battery-electric (BEVs) with fuel economy of 3.6 mi/kWh in the baseline projection by 2040. The grid emissions factor in 2040 is 0.44 kg CO ₂ eq/kWh based on the Xcel Energy reference scenario.	A baseline forecast is based on MA3T data. The Met Council region is disaggregated into a set of smaller geographic regions. Population, vehicle ownership, and other input data are taken from census and travel survey sources. These forecasts use 2015 as a base year for vehicle stocks and sales and compared these results against control totals for 2020. Vehicle electrification changes the vehicle sales and stocks. These changes are then translated into changes in VMT and GHG emissions (direct and indirect) by powertrain type through the basic VMT equation given above.
B2. Phase out biodiesel buses from fleet by 2035 and HEV bus sales by 2050.	Battery-electric buses (BEBs) are more energy efficient than biodiesel buses and will further reduce emissions as the grid decarbonizes. The model assumes that all biodiesel buses are phased out by 2035. The bus fleet is 31% hybrid-electric and 69% battery-electric by 2040.	Changes to the bus fleet produce changes in the powertrain mix of the bus stock. These changes then translate into changes in VMT and GHG emissions (direct and indirect) by powertrain type through the basic VMT equation given above.
C. Introduce dynamic ride-sharing (DRS) in 2025 to replace 90% of motorized PMT by 2040.	Dynamic ride-sharing is a system of shared (autonomous) vehicles that are dispatched to match travelers and pool trips by public or private fleet operators. We assume that average vehicle occupancy is 1.25 and a vehicle can serve 10 travelers per day. DRS vehicles are assumed to have HEV, PHEV, or BEV powertrains. For PHEVs or BEVs, an	DRS is introduced to the tool as a new mode. It draws PMT away from other motorized modes. VMT calculations are based on simulation results in MSP; Austin, TX; and other regions. Assumptions are made about the average VMT per vehicle and number of passengers that can be served by each vehicle. For BEV and PHEV powertrain DRS, VMT is added to account for charging requirements.

**Transportation flow chart for St. Paul
(Passenger)**

	additional 4.5% VMT is incurred for battery recharging.	
D. Introduce autonomous vehicles in 2025 to replace 10% of motorized PMT by 2040.	Autonomous vehicles (AVs) are a technological innovation in the transportation sector on the horizon. They are generally anticipated to increase VMT by making travel more enjoyable and reducing (or eliminating) the driver burden. They are included in the tool to quantify their negative effects, if not appropriately managed.	AV is introduced to the tool as a new mode. It draws PMT away from other motorized modes. VMT calculations are based on simulation results in several US regions. PMT changes are applied based on anticipated increased auto VMT and decreased transit, walking, and cycling travel.
D. Transit improvements		
D1. Increase average vehicle occupancy (AVO) by 10% by 2040.	Increased vehicle ridership decreases the per passenger GHG emissions of transit. We allow the user to specify an increase in AVO to test its effect on transportation GHG emissions.	AVO for each transit mode is adjusted in the basic VMT equation given above.
D2. Increase transit ridership by 10% by 2040.	Increasing transit ridership shifts passenger away from auto trips at the rate of 47 trips per 100 new transit trips.	PMT for each transit mode is adjusted in the basic VMT equation given above.
E. Increase daily fraction of telework by 10 percentage points by 2040.	Telework can reduce motorized travel and the need to work from the office. We estimate a 2.75% decrease in auto VMT for each person-day of telework.	PMT for auto travel is adjusted in the basic VMT equation given literature estimates of VMT reduction for all household travel. This reduction includes commuting and travel induced by additional discretionary time for the worker and additional travel by other household members who now have access to the vehicle. There is less evidence about the effects of telework on transit, walking, and cycling travel. It is generally assumed that the impacts on PMT are negligible or do not have negative environmental impacts (i.e., no significant changes in GHG emissions).
F. Economic instruments		
F1. Introduce a 5 cents per mile vehicle miles traveled (VMT) fee in 2025.	A VMT fee places a price on auto travel with the goal of reducing VMT and shifting travel to other modes. We estimate a 2.0% decrease in auto VMT for a 10% increase in the price of auto travel. We estimate a 1.3%	Price instruments affect VMT via elasticities. The price is measured relative to a marginal cost estimate. The change in VMT by mode is estimated based on literature estimates of both elasticity and marginal cost of travel.

**Transportation flow chart for St. Paul
(Passenger)**

	increase in transit, walking, and cycling VMT for a 10% increase in the price of auto travel. Further, we estimate a 2.5% decrease in freight truck VMT for a 10% increase in the price of truck travel.	The marginal cost used for the VMT fee incorporates the cost of fuel and value of travel time.
F2. Introduce a 5 cents per mile congestion fee in 2025.	Congestion pricing increases the cost of travel at the times of highest demand. Ideally, the price should vary with time of day and location. To simplify implementation, we apply it as a fixed price on an estimated 10.9% of total VMT. We estimate a 1.0% decrease in auto VMT for a 10% increase in the price of congested auto travel. We estimate a 1.3% increase in transit, walking, and cycling VMT for a 10% increase in the price of congested auto travel.	Price instruments affect VMT via elasticities. The price is measured relative to a marginal cost estimate. The change in VMT by mode is estimated based on literature estimates of both elasticity and marginal cost of travel. The marginal cost used for the congestion fee incorporates the cost of fuel and value of travel time.
F3. Introduce a 5 cents per mile pay-as-you-drive (PAYD) insurance in 2025.	PAYD insurance ties the price of insurance to auto VMT. We estimate a 2.0% decrease in auto VMT for a 10% increase in the price of insurance. We estimate a 1.3% increase in transit, walking, and cycling VMT for a 10% increase in the price of insurance.	Price instruments affect VMT via elasticities. The price is measured relative to a marginal cost estimate. The change in VMT by mode is estimated based on literature estimates of both elasticity and marginal cost of travel. The marginal cost used for the PAYD insurance is based on the annual cost of insurance.
F4. Double the gas tax (federal+state) in 2025 (currently 47 cents per gallon).	A gas tax targets low efficiency ICE vehicles and charging a tax per gallon of consumed fuel. We estimate a 1.1% decrease in auto VMT for a 10% increase in the price of gas. The tax applies to travel by gasoline, diesel, HEVs, and PHEVs (the fraction of travel occurring in gasoline-mode).	Price instruments affect VMT via elasticities. The price is measured relative to a marginal cost estimate. The change in VMT by mode is estimated based on literature estimates of both elasticity and marginal cost of travel. The marginal cost used for the gas tax is based on the cost of gasoline at the pump. It is only applied to SI, CI, HEV, and a portion of PHEV VMT since BEVs do not require gasoline.

**Transportation flow chart for St. Paul
(Passenger)**

F5. Double parking pricing beyond current average of 22 cents per hour.	Increasing the price of parking can discourage unnecessary auto trips and encourage the use of transit, walking, and cycling modes. We estimate a 0.7% decrease in auto VMT for a 10% increase in the price of parking. We estimate a 0.1% increase in transit, walking, and cycling VMT for a 10% increase in the price of parking.	Price instruments affect VMT via elasticities. The price is measured relative to a marginal cost estimate. The change in VMT by mode is estimated based on literature estimates of both elasticity and marginal cost of travel. The marginal cost used for the parking price is based on the cost of parking at the trip destination by city taken from travel survey results. Note, in most cases parking has no cost since it will include parking at home, on unmetered streets, and in unpriced parking lots. The marginal cost of parking is assumed to be one cent per hour in most cases to approximate the virtually zero marginal cost of parking.
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